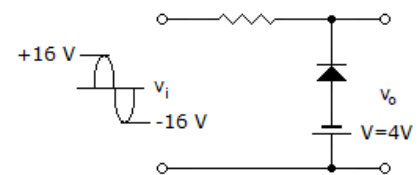


**Multiple Choice:**

1. In the output of a full-wave rectifier:
 - A. The whole wave is inverted.
 - B. The effective value is less than that of the original ac wave.
 - C. The effective value is the same as that of the original ac wave.
 - D. The effective value is more than that of the original ac wave.
2. A simple power supply filter can be made using:
 - A. A capacitor in parallel with the rectifier output.
 - B. A resistor in parallel with the rectifier output.
 - C. An inductor in parallel with the rectifier output.
 - D. A capacitor in series with the rectifier output.
3. Which of the following would make the best filter for a power supply?
 - A. A capacitor in series
 - B. A choke in series
 - C. A capacitor in series and a choke in parallel
 - D. A capacitor in parallel and a choke in series
4. A disadvantage of a half-wave rectifier is the fact that:
 - A. The voltage is excessive
 - B. The current output is low
 - C. The output is hard to filter
 - D. It needs many diodes
5. A current surge takes place when a power supply is first turned on, because:
 - A. The transformer core is suddenly magnetized
 - B. The diodes suddenly start to conduct
 - C. The filter capacitor(s) must be initially charged
 - D. Arcing takes place in the power switch
6. A component that discharges filter capacitors after a power supply has been switched off is called:
 - A. A discharge inductor
 - B. A rectifier diode
 - C. A bleeder resistor
 - D. A filter discharger
7. What is the secondary voltage on a transformer having a primary to secondary turns ratio of 14:1?
 - A. 12.1 V_{rms}
 - B. 15.7 V_{rms}
 - C. 220 V_{rms}
 - D. 85 V_{rms}
8. Which of the following might not be needed in a power supply?
 - A. The transformer
 - B. The filter
 - C. The rectifier
 - D. All of the above are generally needed
9. In a rectifier diode, current flows for approximately how much of the ac cycle?
 - A. 360 degrees
 - B. 270 degrees
 - C. 180 degrees
 - D. 90 degrees
10. The ripple frequency from a full-wave rectifier is:
 - A. twice that from a half-wave circuit
 - B. the same as that from a half-wave circuit
 - C. Half that from a half-wave circuit
 - D. one-fourth that from a half-wave circuit
11. Voltage regulation can be accomplished by a Zener diode connected in:
 - A. Parallel with the filter output, forward-biased
 - B. Parallel with the filter output, reverse-biased
 - C. Series with the filter output, forward-biased
 - D. Series with the filter output, reverse-biased
12. The output from an unfiltered half-wave or full-wave rectifier is a
 - A. pulsating dc voltage.
 - B. steady dc voltage.
 - C. smooth dc voltage.
 - D. None of the above.
13. A voltage regulator with a no-load output dc voltage of 12 V is connected to a load with a resistance of 10 ohms. If the load resistance decreases to 7.5 ohms, the load voltage will decrease to 10.9 V. The load current will be _____ and the percent load regulation is _____.
 - A. 1.45 A, 90.8 %
 - B. 1.45 A, 10.09 %
 - C. 1.2 A, 90.8 %
 - D. 1.2 A, 10.09 %
14. A fuse with a nothing but a straight wire inside is probably:
 - A. A slow-blow type
 - B. A quick-break type
 - C. Of a low current rating
 - D. Of a high current rating
15. The output of a rectifier is:
 - A. 60-Hz ac
 - B. Smooth dc
 - C. Pulsating dc
 - D. 120-Hz ac
16. What transformer turns ratio is required to match a 16-Ω speaker load so that the effective load resistance seen at the primary is 10 kΩ?
 - A. 25:1
 - B. 20:2
 - C. 1:25
 - D. 2:20
17. A simple power supply filter can be built with:
 - A. A capacitor in series with the dc output
 - B. An inductor in parallel with the dc output
 - C. A rectifier in parallel with the dc output



- D. A capacitor in parallel with the dc output
18. Which of the following appliances, which would need the biggest transformer?
- A clock radio
 - A TV broadcast transmitter
 - A shortwave radio receiver
 - A home TV set
19. If you needed exceptionally good ripple filtering for a power supply, the best approach would be to:
- Connect several capacitors in parallel
 - Use a choke-input filter
 - Connect several chokes in series
 - Use two capacitor/choke sections one after the other
20. An advantage of full-wave bridge rectification is:
- It uses the whole transformer secondary for the entire ac input cycle
 - It costs less than other rectifier types
 - It cuts off half of the ac wave cycle
 - It never needs a regulator
21. In a step-down transformer:
- The primary voltage is greater than the secondary voltage
 - The primary impedance is less than the secondary impedance
 - The secondary voltage is greater than the primary voltage
 - The output frequency is higher than the input frequency
22. An autotransformer:
- Works automatically
 - Has a center-tapped secondary
 - Has one tapped winding
 - Is useful only for impedance matching
23. Half-wave rectification means that:
- Half of the ac wave is inverted.
 - Half of the ac wave is chopped off.
 - The whole wave is inverted.
 - The effective value is half the peak value.
24. The shell method of transformer winding:
- Provides maximum coupling
 - Minimizes capacitance between windings
 - Withstands more voltage than other winding methods
 - has windings far apart but along a common axis
25. In a transformer, a center tap would probably be found in:
- The primary winding
 - The secondary winding
 - The unbalanced winding
 - The balanced winding
26. What component is used to smooth out the pulsations across the output of the rectifier?
- inductor
 - diode
 - capacitor
 - regulator
27. Transient suppression minimizes the chance of:
- Diode failure
 - Transformer failure
 - Filter capacitor failure
 - Poor voltage regulation
28. The insertion of a _____ between each of several amplifier stages and a common power supply reduces interaction due to the common power supply impedance.
- high-pass filter
 - band-pass filter
 - low-pass filter
 - band-stop filter
29. In a supply designed to provide high power at low voltage, the best rectifier design would probably be:
- Full wave, center-tap
 - Bridge
 - Voltage multiplier
30. High voltage is used in long-distance power transmission because:
- It is easier to regulate than low voltage
 - The I^2R losses are lower
 - The electromagnetic fields are stronger
 - Smaller transformers can be used
31. What transformer turns ratio is required to match an 8-ohm speaker load so that the effective load resistance seen at the primary is 12.8 kilohms?
- 20:1
 - 50:1
 - 40:1
 - 60:1
32. The part of a power supply immediately preceding the regulator is:
- The transformer
 - The rectifier
 - The filter
 - The ac input
33. If there is reactance at the output of an impedance transformer:
- The circuit will not work
 - There will be an impedance mismatch, no matter what the turns ratio of the transformer
 - A center tap must be used at the secondary
 - The turns ratio must be changed to obtain a match
34. Determine the peak for both half cycles of the output waveform.



- A. 16 V, -4 V



- B. 16 V, 4 V
C. -16 V, 4 V
D. -16 V, -4 V
35. When _____ are used to produce the d.c. supplies of electronic systems, a large ripple introduces undesirable noise and may even mask the effect of the signal voltage.
- A. transformer
B. rectifier
C. filter
D. regulator
36. If a fuse blows, and it is replaced with one having a lower current rating, there's a good chance that:
- A. The power supply will be severely damaged
B. The diodes will not rectify
C. The fuse will blow out right away
D. Transient suppressors won't work
37. A transformer has a secondary-to-primary turns ratio of 0.167. This transformer is:
- A. A step-up unit
B. A step-down unit
C. Neither step-up nor step-down
D. A reversible unit
38. When rectifiers are used to produce the d.c. supplies of electronic systems, a large ripple introduces undesirable noise and may even mask the effect of the signal voltage. What is added to smooth the output voltage?
- A. High-pass filter
B. Band-pass filter
C. Low-pass filter
D. Band-stop filter
39. In a step-up transformer:
- A. The primary impedance is greater than the secondary impedance
B. The secondary winding is right on top of the primary
C. The primary voltage is less than the secondary voltage
D. All of the above
40. The principal disadvantage of a voltage multiplier is:
- A. Excessive current
B. Excessive voltage
C. Insufficient rectification
D. Poor regulation
41. In the output of a half-wave rectifier:
- A. Half of the wave is inverted.
B. The effective value is less than that of the original ac wave.
C. The effective value is the same as that of the original ac wave.
D. The effective value is more than that of the original ac wave.
42. Calculate the dc voltage across a 1-kohm load for an RC filter section ($R=120\ \Omega$, $C=10\ \mu\text{F}$). The dc voltage across the initial filter capacitor is $V_{dc} = 60\text{ V}$.
- A. 45.5 V C. 47.82 V
B. 51.77 V D. 53.6 V
43. Eddy currents cause:
- A. An increase in efficiency
B. An increase in coupling between windings
C. An increase in core loss
D. An increase in usable frequency range
44. Bleeder resistors are:
- A. Connected in parallel with filter capacitors
B. Of low ohmic value
C. Effective for transient suppression
D. Effective for surge suppression
45. How many diodes are used to obtain full-wave rectification where a center-tapped transformer T is used?
- A. 1 C. 2
B. 3 D. 4
46. The capacitance between the primary and the secondary windings of a transformer can be minimized by:
- A. Placing the windings on opposite sides of a toroidal core.
B. Winding the secondary right on top of the primary
C. Using the highest possible frequency
D. Using a center tap on the balanced winding
47. A full-wave rectified signal of $18V_p$ is connected to a $400\ \mu\text{F}$ filter capacitor. What is the dc voltage across the capacitor at a load of 100mA ?
- A. 17.35 V C. 18.24 V
B. 16.96 V D. 19.90 V
48. The variation in potential between points on a rectified output is known as:
- A. effective C. ripple
B. alternation D. e.m.f
49. With a 200 V peak value of a half-wave rectifier, what would be the average value?
- A. 63.6 V C. 6.36 V
B. 3.56 V D. 5.12 V
50. To service a power supply with which you are not completely familiar, you should:
- A. Install bleeder resistor
B. Use proper fusing
C. Leave it alone and have a professional work on it
D. Use a voltage regulator
- END